

Data job market analysis, India

Which skills data roles ask for in India, how demand moves, and where demand and pay line up.

TYPE Personal project	TOOLS Python, pandas, Matplotlib, Seaborn, Jupyter	DATA data_jobs (Luke Barousse), India
BUILT Nov 2024		

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The question

Which skills do data roles ask for in India, how is that demand changing, and which skills combine strong demand with higher pay?

The project answers four questions: the most requested skills for the three most common data roles; how skill demand trends for Data Analysts; how jobs and skills pay; and which skills are the best to learn.

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The data

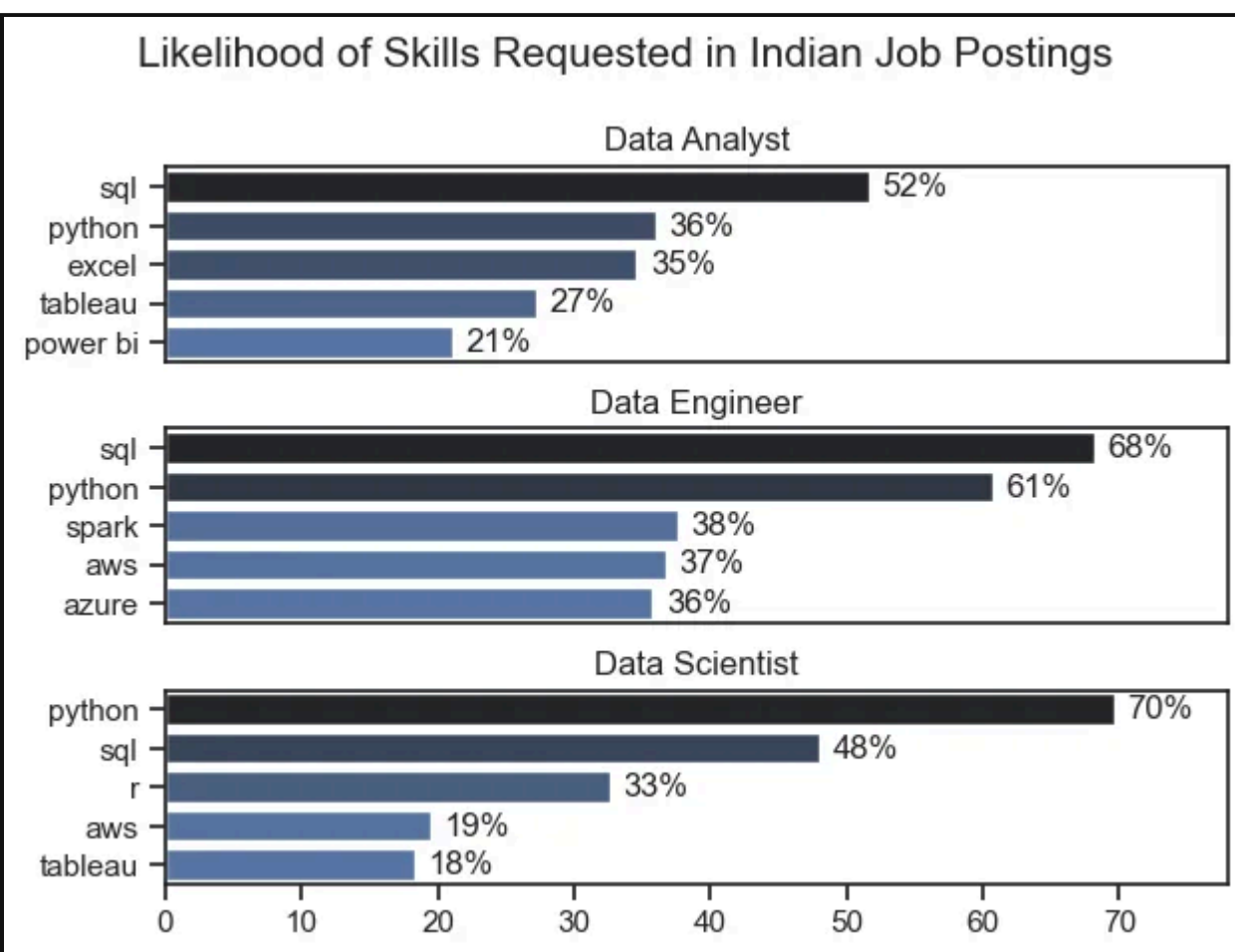
The data_jobs dataset from Luke Barousse's Python course: job postings with titles, locations, salaries and skills. I narrowed it to postings in India.

The method

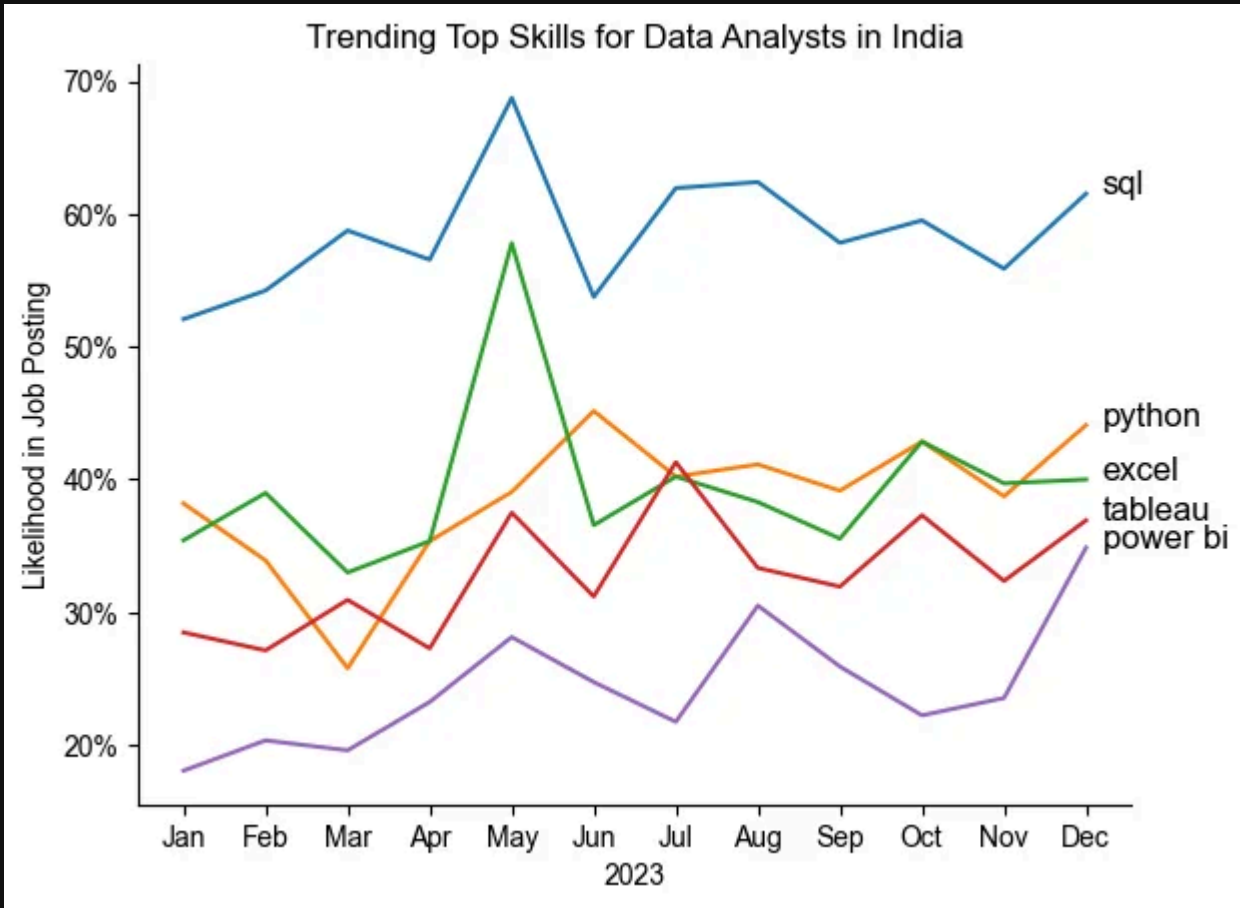
pandas to clean dates and skill lists, filter to India and group by role, skill and month; Matplotlib and Seaborn for the charts. Five Jupyter notebooks walk through the steps, from an exploratory intro to the optimal-skills view.

What I found

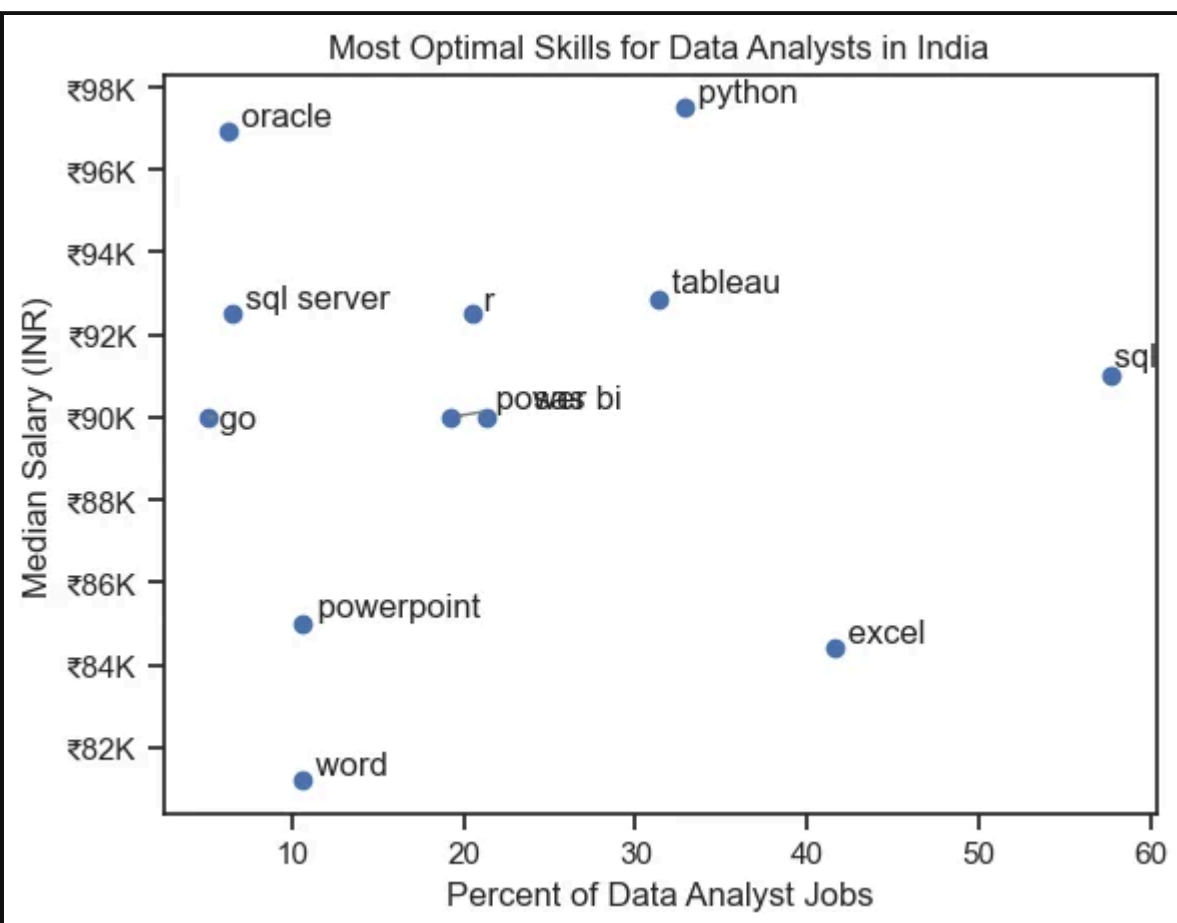
- **SQL leads.** It is the most requested skill for Data Analysts and Data Scientists (in over half of postings), and appears in 68% of Data Engineer postings. Python matters most for Data Scientists (70%) and Data Engineers (61%).
- **Excel demand rose through 2023.** From around May it climbed past Python and Tableau by year end, while SQL slipped slightly.
- **Demand and pay are different lists.** Specialised skills such as PostgreSQL, PySpark and Linux went with higher pay; Power BI, Spark and Tableau were among the most requested.



Share of postings requesting each skill, by role.



Monthly demand for the top Data Analyst skills, 2023.



Skills by demand and median pay.

What I'd do next

- Report how many postings sit behind every salary figure; pay data in job ads is often sparse, so small groups should be flagged.
- Compare India with other markets using the same code.
- Look at skill combinations, not single skills, since postings ask for sets.

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Links

- [Notebooks, charts and write-up on GitHub](https://github.com/samriddhosaha/Python-Projects/tree/main/Data-Job-Market-Analysis) ↗
- [Luke Barousse's Python course \(data source\)](https://lukebarousse.com/python) ↗

Last updated 6 Oct 2026.

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